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function build_simulink_model()
%BUILD_SIMULINK_MODEL Builds an executable Simulink harness for MATLAB R2021a.

params = initialize_paper_parameters();
modelName = 'uav_landing_mpc_faa_model';

if bdIsLoaded(modelName)
    close_system(modelName, 0);
end
new_system(modelName);
open_system(modelName);

add_block('simulink/Sources/Clock', [modelName '/Clock'], 'Position', [40 90 70 120]);
add_block('simulink/User-Defined Functions/Interpreted MATLAB Function', ...
    [modelName '/MPC FAA Landing Step'], 'Position', [150 70 370 140]);
add_block('simulink/Sinks/To Workspace', [modelName '/landing_logs'], 'Position', [470 85 570 125]);

set_param([modelName '/MPC FAA Landing Step'], 'MATLABFcn', 'simulink_landing_step');
set_param([modelName '/landing_logs'], 'VariableName', 'simulink_landing_logs');
set_param([modelName '/landing_logs'], 'SaveFormat', 'Array');

add_line(modelName, 'Clock/1', 'MPC FAA Landing Step/1');
add_line(modelName, 'MPC FAA Landing Step/1', 'landing_logs/1');

set_param(modelName, 'StopTime', num2str(params.stateMachine.maxTime));
set_param(modelName, 'Solver', 'FixedStepDiscrete');
set_param(modelName, 'FixedStep', num2str(params.sim.dt));

annotationText = sprintf(['This Simulink harness mirrors the paper pipeline:\n' ...
    'USV prediction -> mission state machine -> MPC-FAA trajectory generator -> linearized UAV plant.\n' ...
    'The Interpreted MATLAB Function calls simulink_landing_step.m, which uses the same functions as run_main.m.']);
note = Simulink.Annotation(modelName, annotationText);
note.Position = [40 300 760 370];

save_system(modelName, fullfile(fileparts(mfilename('fullpath'))), [modelName '.slx']);
fprintf('Created Simulink model: %s\n', fullfile(fileparts(mfilename('fullpath'))), [modelName '.slx']);
end

```

Warning: The model name 'uav_landing_mpc_faa_model' is shadowing another name in the MATLAB workspace or path. Type "which -all uav_landing_mpc_faa_model" at the command line to find the other uses of this name. You should change the name of the model to avoid problems.

Created Simulink model: /Users/jamiuolaideoyetunde/Desktop/Research Internship - ITMO Course Work/UAV_Landing_MPC_ReverseEngineering/matlab/uav_la

